Suppose that the swap curve is as follows:

Maturity	Swap Rate
1 year	3.00%
2 year	3.25%
3 year	3.50%
4 year	3.75%
5 year	4.00%

These rates are annual bond versus 1-year LIBOR. Suppose further that the lognormal swaption matrix prevailing at this time is as given in Table 5.1.

Suppose that a swap dealer sells \$100mm 1 into 3 swaption straddles struck at-the-money forward to one of its clients.

a) Assuming that the dealer is willing to sell the straddles at mid-market, what mid-market lognormal swaption vol should be used to price the trade? What is the premium at which the straddles should be sold if the dealer uses the lognormal model to price options and uses this vol as an input?

b) Given the implied vol from part a), what is the equivalent normal implied vol? What is the premium at which the straddles should be sold if the dealer uses the normal model to price options and uses this vol as an input?

c) Referring to part a), after the dealer sells the straddles suppose that he wishes to maintain a delta-neutral position by trading in the underlying 1x4 swap. What position in the underlying 1x4 swap should the dealer establish if he wishes to be delta-neutral, assuming he uses the lognormal model to price and hedge? Express your answer in the notional of 1x4 swaps that he should enter. Be sure to indicate if the dealer should pay or receive on this position of 1x4 swaps.

d) Referring to part a), after the dealer sells the straddles suppose that he wishes to maintain a delta-neutral position by trading in the underlying 1x4 swap. What position in the underlying 1x4 swap should the dealer establish if he wishes to be delta-neutral assuming he uses the normal model to price and hedge? Express your answer in the notional of 1x4 swaps that he should enter. Be sure to indicate if the dealer should pay or receive on this position of 1x4 swaps.

e) Now consider a dealer who sells \$100mm of a 1 into 3 swaption straddle struck at $X\%$. Suppose that the dealer wants to remain delta neutral. How much of the underlying 1x4 swap should the dealer enter into? Show a graph that demonstrates the appropriate hedge, assuming the dealer uses the normal model and also assuming the dealer uses the lognormal model, allowing X to vary between 2% and 6%. Be sure to indicate if the dealer should be paying fixed or receiving fixed in the underlying 1x4 swap for each of the strikes you consider.

Chapter 6

Swaps with Embedded Options

In this chapter we will take a look at a variety of swaps that have one or more embedded options in them. Typically, buy-side investors receive fixed in such swaps and short the options that are being embedded into the trade. As we will see, this results in a trade in which the investor receives a higher fixed coupon than he would otherwise get in a plain vanilla swap.

Oftentimes these trades are geared in such a way that they provide the highest fixed rate possible to the investor. In the swaps market, such trades are sometimes referred to as yield enhancing trades, and investors have used such trades as surrogate investment opportunities. Of course, the high coupons often obtainable through such trades come at a price: the investor has sold options implicit in these structures in order to achieve a high coupon. When those options move in-the-money, that has negative consequences for the investor.

6.1 An Underlying Concept

As with almost every swap that we will examine, any swap with embedded options that we will look at here is an NPV = 0 transaction when first executed. The fixed rate in the trade will be set in a way such that the trade is worth zero to both counterparties when transacted. Since the investor is usually receiving fixed from and selling options to the dealer in most of these structures, this implies that the fixed rate coupon that the investor receives will be higher than the par swap rate in a plain vanilla swap, so as to compensate the investor for the short option position in the trade. Again, the fixed coupon in such a swap, which we will denote by x, is set such that

$$\text{NPV of Swap with Embedded Options} = 0 \tag{6.1}$$

when the trade is first executed.[1] It will often be useful to conceptualize such trades by bifurcating them into their component parts: a plain vanilla fixed-floating swap with a fixed coupon of x, and some options. With this type of thinking, we can then write that

$$\begin{aligned}&\text{Value of Plain Vanilla Swap with Coupon of } x\\&+ \text{Value of Embedded Options}\\&= \;0,\end{aligned} \tag{6.2}$$

when the trade is first executed. Let's think about Equation 6.2 from the point of view of the dealer who is paying fixed in such a trade. The value of the embedded options that the dealer is buying is a positive number. This implies that the coupon, x, will be set in the trade such that the value of the plain vanilla swap is negative to the dealer.[2] By extension, the (off-market) fixed rate x will be above the par swap rate for the given maturity swap. In many of the particular trades that we will examine, we will have to iterate to find the fixed rate x that makes the swap with embedded options an NPV $= 0$ transaction.

We already looked at one example of a swap with an embedded option in Section 4.1.3 of Chapter 4 (and you looked at a bunch more in the fifth question of Problem Set 4). In Section 4.1.3 we looked at a trade in which a cap was embedded into a swap. In this particular case, the investor actually paid fixed (whereas in most of the trades considered in this chapter the investor receives fixed) and sold the embedded cap in order to lower the fixed rate he had to pay. Despite the fact that the investor is paying fixed in this swap, Equation 6.2 is still applicable and offers us a way to think about this trade. The terms of this trade had the client paying a (below-market) fixed rate of 1.339% in exchange for receiving the minimum of LIBOR and 3%. The dealer was buying a LIBOR cap struck at 3%, which was worth \$1,470,000. From the perspective of the dealer, this option that the dealer is long represents a positive amount in Equation 6.2. And the value of a plain vanilla swap with a (below-market) coupon of 1.339% when the par rate in a

[1] Of course, the value of these trades will in general move away from zero with the passage of time and the movement of rates, vol, and so on.

[2] Equivalently, looking at this trade from the point of view of the investor who is receiving fixed, the options have been shorted and are of negative value to the investor and the plain vanilla swap with the high coupon x is of positive value to the investor.

5-year swap was 1.647% was found to be $-\$1{,}470{,}000$ from the perspective of the dealer. Thus, in this trade we solved for a coupon $x = 1.339\%$ such that the negative value of the off-market swap was exactly offset by the positive value of the cap that the dealer got long.

6.2 Cancelable Swaps

A (European style) *cancelable swap* is a swap in which the fixed rate payer has the right to terminate the trade at some point in time prior to its stated maturity. For example, consider a "5 noncall 1" (which is also written as "5nc1"), a 5-year swap in which the fixed rate payer has the right to cancel the swap after one year. If the fixed rate payer does cancel the swap after one year, then both counterparties will make final coupon payments at the end of the first year, and no further cash flows will be made in the swap.[3] If the fixed rate payer does not cancel the trade at the end of the first year, then the swap will remain outstanding for the remaining four years. By the way, when will the fixed rate payer decide to cancel the swap?[4]

A cancelable swap, like a regular swap, is an NPV = 0 transaction at inception. The fixed coupon in the swap will be set such that the swap has zero value when it is transacted. The fixed rate will be adjusted from the plain vanilla 5-year swap rate in order to reflect the value of the option that the fixed rate payer is long. In fact, it is clear that the fixed rate in the 5nc1 must be greater than the 5-year swap rate: the fixed rate payer in the 5nc1 will be paying fixed for five years *and* has the option to cancel the trade. This option has value to the fixed rate payer, and thus in order to compensate the floating rate payer for selling this option, the fixed rate must be set higher than the plain vanilla 5-year swap rate. The augmentation of the fixed rate serves to exactly offset the value of the option that the fixed rate payer is long, as we had discussed in Section 6.1.

[3] Note that this is fundamentally different from an unwind of a swap. In an unwind, one counterparty is paying the other to get out of the remaining obligations of the trade, and the unwind fee represents the NPV of the remaining cash flows. In a cancelable swap, the fixed rate payer has the right to cancel at some point and will pay no fee in the event of cancelation.

[4] If, in one year, 4-year swap rates are less than the fixed coupon in the swap, the fixed rate payer will rationally decide to cancel. For instance, suppose that the fixed coupon in the 5nc1 is 6%, and in one year from now 4-year swap rates are 5.50%. The fixed rate payer would choose to cancel. If for some reason he still wanted to pay fixed for four years, he could always do so just by going into the market and paying fixed at 5.50% for four years rather than by paying fixed at 6% in the existing swap, saving himself 50 bps running.

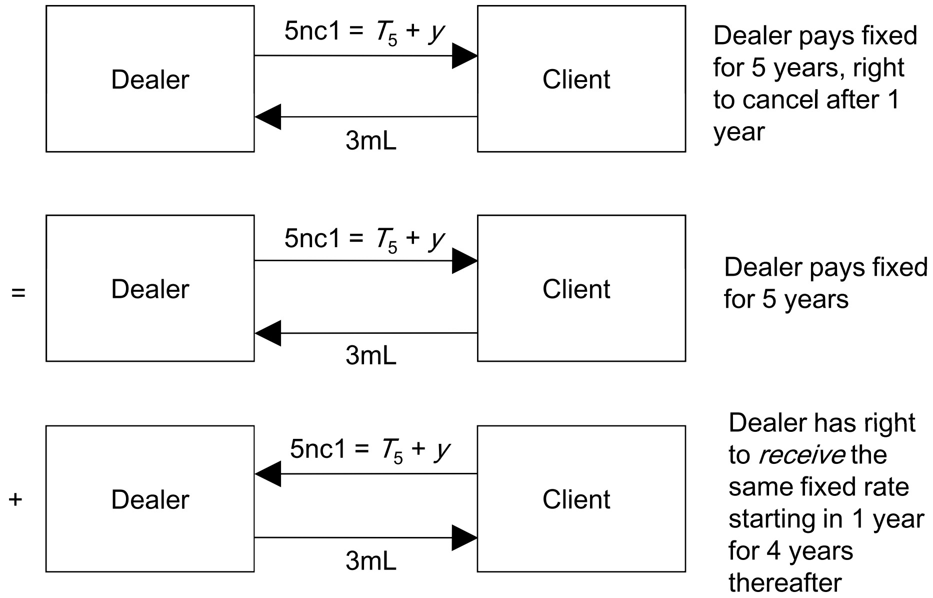

Figure 6.1: Decomposing a Cancelable Swap into a Plain Vanilla Swap and a Receiver Swaption

We can think of breaking a 5nc1 up into two component parts: a plain vanilla 5-year swap and an option to cancel the swap. This is demonstrated in Figure 6.1. The top portion of this figure represents the cancelable swap. The fixed coupon in this trade, denoted 5nc1, will be equal to the yield on the 5-year Treasury, T_5, plus some spread, y. The middle and bottom portions of this figure break the cancelable swap into two components. The middle part is a plain vanilla swap with the same coupon $T_5 + y$, and the bottom part represents an option to cancel the swap after one year. Note the following important point: the option to cancel the swap after one year is equivalent to the dealer having the right to receive that same fixed rate $T_5 + y$ and pay LIBOR for four years starting in one year. Indeed, if this option is exercised, then all cash flows resulting from exercise will exactly net out the remaining cash flows in the plain vanilla swap, and thus the swap will be canceled. This option, in which the dealer has the right to receive a fixed rate for four years starting in one year is of course a 1 into 4 receiver swaption struck at $T_5 + y$.

In a cancelable swap, it is often the dealer who is the fixed rate payer and who therefore owns the right to cancel. Since the dealer is going to pay a fixed rate that is greater than the 5-year swap rate, the plain vanilla swap represented in the middle portion of Figure 6.1 will have negative value to the dealer. The receiver swaption will have positive value to the dealer. When we solve for the cancelable swap rate $T_5 + y$, that rate will be such that the positive value of the swaption will exactly offset the negative value of the plain vanilla swap. This results in a cancelable swap that is an NPV = 0 transaction, as per Equation 6.2.

Investors who receive fixed in cancelable swaps enjoy earning a higher fixed rate (and therefore greater carry) than in a comparable maturity *bullet* (i.e., noncancelable) swap. Of course, this enhanced fixed rate comes at a price: the swap will be canceled in a low rate environment. A fixed rate receiver in a cancelable swap whose swap is canceled will face the prospect of lower rates, should he wish to enter into a new swap.

6.2.1 Some Uses of Cancelable Swaps

Suppose a U.S. agency has decided that it wants to issue debt. The agency explores the possibility of issuing a fixed rate bond, a floating rate bond, and a fixed-rate callable bond and determines that investor appetite is strongest for the callable bond. Callable bonds will pay the bondholder a higher fixed coupon than comparable maturity bullet bonds and as such are often attractive to investors.[5] Among all callable bonds, the agency further determines that investors will have the greatest interest if the agency were to issue a 5-year bond, callable after one year. However, the agency prefers to fund itself in accordance with its floating rate benchmark, which is LIBOR. As shown in Figure 6.2, the agency can synthetically create floating rate funding by issuing the callable bond and entering into a cancelable swap.

Upon issuing the bond and entering into the cancelable swap, the agency has locked in synthetic floating rate funding of 3-month LIBOR $-\,10$ bps over the first year. Issuers of callable bonds who enter into cancelable swaps to obtain synthetic floating rate funding will often take their cue to cancel the bond from the dealer. That is, if the dealer cancels the swap, then the issuer will likely choose to cancel the bond.[6] If after one year the dealer

[5]Of course, that higher fixed coupon comes at a price: callable bonds will generally be called in a low rate environment. If the bond is called, the bondholder will get his principal back and be left with the prospect of having to reinvest his funds in a lower rate environment.

[6]It has historically been common for issuers to structure their cancelable swaps such that the dealer must inform of his intention to cancel the swap some number of days prior to the issuer needing to inform the bondholders of a decision to call. In this way, the issuer upon observing the decision of the dealer regarding the swap can decide what it wants to

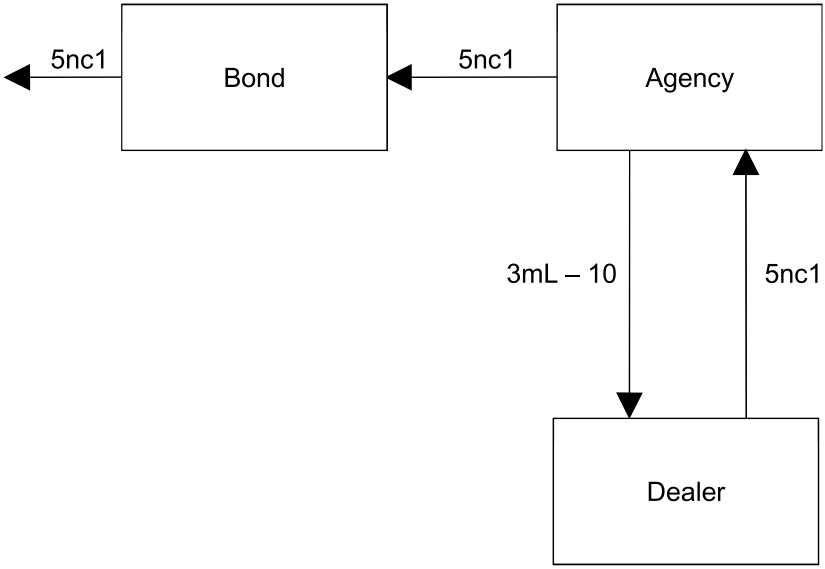

Figure 6.2: Agency Swapping Out Callable Debt Back to Floating

cancels the swap and the agency calls the bond, principal is returned to the bondholders, and there are no further cash flows associated with the swap. If instead the dealer decides not to cancel after one year and the agency does not call the bonds, the agency will continue to obtain synthetic floating rate funding of 3-month LIBOR − 10 for the remaining four years.

6.2.2 Solving for the Fixed Rate in Cancelable Swaps

As mentioned above, the fixed rate in a cancelable swap is set such that the swap is an NPV = 0 transaction. The process of finding the fixed rate such that the value of the trade is zero today is an iterative one. In order to demonstrate this, let us again consider our stylized example, with the curve as shown in Table 6.1, and consider solving for the fixed coupon in a 4nc1.

do about calling the bond. Although issuers often link their decision to call the bond to the dealer's decision to cancel the swap, this is not to say that the decision of the issuer cannot deviate from the decision of the dealer.

Maturity	Treasury Yield	Swap Spread	Swap Rate (Ann 30/360)
1	4.80%	20	5.00%
2	4.88%	23	5.11%
3	5.05%	25	5.30%
4	5.19%	28	5.47%
5	5.25%	35	5.60%

Table 6.1: The Swap Curve

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	(54,700)	(52,095)	50,000	47,619
2	5.113%	5.226%	(54,700)	(49,508)	52,257	47,297
3	5.312%	5.712%	(54,700)	(46,833)	57,122	48,907
4	5.494%	6.041%	(54,700)	(44,165)	60,414	48,778
				(192,601)		192,601
		Fixed rate	5.470%	NPV	0	
		Notional	1,000,000			

Figure 6.3: Calculating the 4-Year Swap Rate

A 4nc1 is a 4-year swap in which the fixed rate payer has the right to cancel the trade after one year. We will assume that the dealer is the one who is paying fixed and has the right to cancel. We are going to follow an iterative procedure in order to solve for the cancelable swap rate. Let's start out by assuming the cancelable swap rate is equal to the 4-year swap rate. We know this will be wrong since the 4nc1 fixed rate must be higher than the 4-year swap rate, but we have to start somewhere. The curve in Table 6.1 indicates that the 4-year swap rate is 5.47%, and as shown in Figure 6.3 we indeed obtain this rate when we bootstrap the curve. Note that in this figure we have accentuated the dealer's position by signing the cash flows (negative cash flows indicate what the dealer is paying out, and positive cash flows indicate what the dealer is receiving).

So this is our initial guess for the 4nc1 fixed rate: the plain vanilla 4-year swap rate. Again, we know that this is not going to be the answer because the 4nc1 rate is going to have to be higher than the 4-year swap rate. Now, if 5.47% were the cancelable swap rate, that would imply that the dealer would be long a 1 into 3 receiver swaption struck at 5.47%.

Now let's use Black's formula as given in Equation 5.16 to price up this 1 into 3 receiver swaption struck at 5.47%.[7] This calculation is shown in the top half of Figure 6.4, using an assumed implied volatility of 20%. The value of the 5.47% strike receiver is 92.7 bps of the notional, or $9,271 on the assumed notional of $1mm. In the bottom half of Figure 6.4, we incorporate the value of this receiver into our cancelable swap model. Note that from the dealer's perspective he is paying out fixed coupons of 5.47%, and he is long the receiver swaption. The swap is worth nothing (5.47% is the par swap rate), and the receiver is worth $9,271 in his favor. Thus, as expected, 5.47% is not the cancelable swap rate because it is not that fixed coupon that makes the cancelable swap an NPV = 0 transaction.

In order to iterate to the solution, we now raise the fixed coupon in our cancelable swap model until the point where the negative value of the swap offsets the positive value of this receiver swaption. Figure 6.5 shows that this occurs when the fixed rate is set equal to 5.733%. So does this mean that 5.733% is the cancelable swap rate? No, it doesn't. Although we have raised the fixed rate in our cancelable swap model in Figure 6.5 such that the net of the value of the swap and the value of the swaption is zero, we need to be careful here. The value of the swaption is for a receiver with a strike of 5.47%, not 5.733%. So we now need to revalue the swaption with a strike of 5.733%, which is shown in the top half of Figure 6.6. The value of the 5.733% strike receiver is $12,819, and in the bottom half of Figure 6.6 we incorporate this value into our cancelable swap model. This figure shows that the net of the swap with a 5.733% fixed coupon and the receiver swaption struck at 5.733% does not equal zero. Therefore, we must again raise the fixed coupon until the net value of the swap and the receiver swaption is equal to zero. Figure 6.7 shows that this occurs when we set the fixed rate to 5.834%. But of course the value of the swaption in Figure 6.7 is for a receiver with a strike of 5.733%, not 5.834%, and thus we need to revalue the swaption with a strike of 5.834%. And so on, and so on, and so on. We continue to iterate until we stop needing to move the fixed coupon any higher. This will occur when the off-marketness of the fixed coupon exactly offsets the value of the

[7] Despite the discussion in Chapter 5 indicating that at this point the normal model has eclipsed Black's model as the market standard, we will continue to price options using Black's model, and we will assume flat skew (i.e., the lognormal implied vol does not vary with strike).